

Business case for Amazon's flying warehouse

Amazon being the largest online retail shop that maintains warehouses of products from various sellers across geo locations, improving customer experience and turnaround time for delivery of products have been matters of great concern for them. They have conceptualised the flying warehouse which has UAVs deployed for product delivery.

Both Amazon and Walmart have been researching a lot since a decade on improving their customer experience in delivering products to remote locations and simplify product delivery based on various factors including frequently ordered items, urgency in critical orders, and many more. To handle this kind of customer experience, flying warehouse is being experimented and Amazon has received a series of patents for it in 2016 and is making a heavy investment in deployment of AFCs and the UAVs.

Implementation design of Amazon's flying warehouse

Amazon flying warehouse was conceptualised a decade ago and implementation of the draft design started in 2013 and later matured to its final design in 2016 when it was patented. In August 2020, Amazon got official Federal Aviation Administration (FAA) approval to use drones for medical supply delivery.

The important components in the conceptual design of Amazon flying warehouse, shown in Fig. 2, include the following.

Airborne fulfilment centre (AFC).

This is the flying warehouse, a costly investment, to be built in the sky nearest to the delivery sites. It would be used for storage of products received from a ground warehouse and as parking/flying space for drones. AFC will also have technicians for repairs and maintenance and a support team to handle the packages. It would be a one-time investment of varying size depending on the products to be handled.

Unmanned aerial vehicles (UAVs). These are small transporting vehicles or carriers that would lift products to deliver as directed by



Fig. 3: Amazon's AFC vehicles (Image source: Amazon press release)

the UAV delivery team. UAVs will be GPS enabled and move as per path programmed for their delivery. Usually, a delivery location or city will be closer to 80 kilometres (50 miles) and the UAV would be expected to deliver within an hour in most cases. UAVs will be tracked and monitored by the support team in AFC to manage the delivery.

Amazon ground warehouse. This is the regular Amazon warehouse where all products are stored by various vendors. Select items from this ground warehouse will be taken to AFC acting as a temporary delivery store for the nearest delivery locations. A single Amazon ground warehouse can be used to provide products for multiple AFC locations.

Replenishment shuttle. It is a carrier flight that takes goods from a ground warehouse to an AFC for delivery later on receipt of order from the nearest locations. Replenishment shuttle will fly as per schedule like once in a week. Any inventory requirement on day-to-day basis will be monitored in ground warehouse and considered for order fulfilment through replenishment shuttle in the next trip.

Future trends

Amazon warehouse is experimenting with medical supply delivery as there is a huge demand to supply urgent medical requirements like oxygen cans, emergency injections, and drugs. UAVs can deliver quickly even in remote locations, which is important in case of emergencies.

Walmart is also experimenting with its own flying warehouse concept for quickly delivering daily supplies like pantry items to its premium regular customers.

We can expect Amazon flying warehouse targeting delivery of daily supplies like food and drinks next, after medical supplies, for its regular customers on subscription basis (say, daily delivery to customers with monthly payment from prime customers). As per an article on Amazon flying warehouse on *futurism.com*, we can expect retail industry to use this disruptive technology to realign supply-chain process. As a result, small retail shops in large cities could get replaced with UAV based delivery in next ten years.

Also, we can expect a cloud of AFCs to be set up across multiple geo-locations in a country (like edge locations) to fulfil and share product delivery across locations. This will enable faster delivery and ensure product availability based on automatic replenishment of the products in demand. **EFY**

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